When Does Learning How to Ask Cost More in Interactive Recommendation?

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ABSTRACT

An online recommender may need to learn both a visitor's preferences and which question formats yield reliable answers. Does learning how to ask necessarily require extra interaction? We study best-item identification in a finite affine catalogue with unknown binary preferences, group-shared better formats, and two unknown response-reliability levels. Relative to an oracle knowing only the better formats, we characterize exactly when their ignorance increases the leading query-cost coefficient. The boundary depends jointly on decision-changing preference sets and reliability sharing, and is independent of the worse reliability level without calibration. If the oracle's coordinate allocation is unique and unsaturated, the premium vanishes exactly when its support lies in one sharing group. An explicit catalogue gives opposite outcomes for partitions with identical group sizes. We instantiate joint inference and decision-directed acquisition with an anytime recommendation certificate. Controlled mechanism tests illustrate the model and retain strong information-gain comparisons; they do not establish superiority or natural-user validity.

CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

Interactive recommendation, preference elicitation, unknown reliability, fixed confidence

1 INTRODUCTION

A shopping assistant can learn a useful way to ask a question and still gain little from reusing it. Asking about a concrete use case may help a visitor express a requirement that a technical attribute question obscures [9]. Reuse helps only if the remaining requirements matter for the products still under consideration. A platform therefore faces two linked questions: which preferences could change its recommendation, and where does experience with a reliable question transfer?

This is an interaction-cost problem for Web recommendation. An assistant should identify a suitable item without requiring a complete, repeatedly verified user profile. Bayesian preference elicitation already directs questions toward useful information [6]; recent cold-start personalization exploits correlations between preferences [1]. Catalogue-dependent conversational efficiency also has a theoretical foundation [4]. Our question concerns a different source of burden: *the additional evidence required because the reliable observation format is unknown.* Styles can affect conversational recommendation [8], but such findings do not establish the statistical channel assumed here.

We compare a learner with an information-matched oracle: both lack the preferences and response accuracies, while only the oracle knows the better format in each group. This isolates the cost of learning how to ask. Recovering every preference would force the learner to resolve all coordinates; identifying one best item instead requires separating only preference combinations that change the winner. That distinction changes the boundary for whether format learning costs extra.

Learning the reliability of multiple information providers is established in proactive learning [11]; unknown variances can also increase best-arm identification cost [7]. Our contribution is an exact zero-versus-positive boundary for the shared-format problem, not the existence of nuisance-parameter costs. The characterization eliminates alternative reliability parameters, yields an interpretable criterion when the oracle allocation is unique, and separates equally sized group partitions. We instantiate it with a certified acquisition loop. The generic KL framework is established [2, 3, 5]; the new result links recommendation decisions to reliability sharing.

2 RELIABLE BEST-ITEM ELICITATION

Decision and observations. A known finite catalogue has utilities $U_i(\theta) = b_i + v_i^\top \theta$, with fixed unknown $\theta \in \{-1,+1\}^d$. Assume a unique best item $f(\theta)$ at every sign vector and a nonconstant decision map. A known partition $S_1, \dots, S_G$ groups preferences whose more reliable question format $z_g \in \{0,1\}$ is shared. A unit-cost query (j,m) asks about the same preference in format m, returning

$$\Pr_h(Y = \theta_j \mid j,m) = \begin{cases} p, & m = z_{g(j)}, \\ q, & m \neq z_{g(j)}, \end{cases} \qquad \tfrac{1}{2} \le q < p \le 1-\eta, \tag{1}$$

for known $\eta > 0$. The rates are unknown and common across groups; responses are fresh and conditionally independent. Groups are given, not learned. We include no calibration, direct format reports or whole-profile disclosures. The two formats must preserve the underlying decision and access to evidence. These are modelling assumptions, not empirical properties of conversational users.

A matched information benchmark. A policy is uniformly δ-correct if it terminates and returns $f(\theta)$ with probability at least $1-\delta$ for every allowed $h = (\theta, z, p, q)$. Write P_h^e for the Bernoulli law of action $e = (j,m)$, and let $\mathrm{Alt}(h)$ contain states with a different best item. Define

$$K = \min_{x \ge 0} \left\{ \sum_e x_e : \inf_{h' \in \mathrm{Alt}(h)} \sum_e x_e \, \mathrm{kl}(P_h^e, P_{h'}^e) \ge 1 \right\}. \tag{2}$$

Alternative rates use the closure $1/2 \le q' \le p' \le 1-\eta$. The oracle coefficient K_{or} additionally fixes $z' = z$; its rates remain unknown. These are pointwise coefficients at fixed dimension and fixed true rates.

Proposition 1 (Information benchmark). *Every uniformly δ-correct policy satisfies* $\liminf_{\delta\to 0} \mathbb{E}_h\tau/\log(1/\delta) \geq K$. *Ideal forced-sample joint estimation and information tracking with vanishing actual approximation errors attain equality; the oracle analog holds for* K_{or}.

Proof sketch. Stopped-transcript change of measure against each wrong-item state yields (2). For attainment, force $\Omega(\sqrt{t})$ samples of each action, estimate signs, types and rates from forced data, and track the estimated optimizer set with $o(t)$ discrepancy. The all-task direction supplies a locally strict feasibility repair. Consistency and uniform likelihood concentration give $\Pr_h(\tau > t) \leq Ct^a e^{-c\sqrt{t}}$ beyond $(K+\epsilon)\log(1/\delta)$ for sufficiently small δ, with constants depending on the fixed instance. Summing the tail gives the matching upper coefficient. This is the controlled-sensing specialization [3]; fixed solver tolerances do not supply its approximation contract. □

Let θ^F reverse coordinates in F. The minimal decision-changing sets

$$\mathcal{H} = \min_{\subseteq}\{F \subseteq [d] : f(\theta^F) \neq f(\theta)\} \tag{3}$$

are the alternatives relevant to (2). Indeed, every wrong decision contains a minimal such set, and deleting other reversals weakly reduces KL: for $r, x \geq 1/2$, $\mathrm{kl}(r,1-x)-\mathrm{kl}(r,x) = (2r-1)\log(x/(1-x)) \geq 0$. For $w \in \Delta_d$ put $w(F) = \sum_{j\in F} w_j$ and

$$\begin{aligned} \rho &= \max_{w\in\Delta_d} \min_{F\in\mathcal{H}} w(F), \\ \psi_p(s) &= H_b\Big(\max\{\tfrac{1}{2}, p-(2p-1)s\}\Big) - H_b(p). \end{aligned} \tag{4}$$

Here H_b is binary entropy in nats. The reciprocal cover value $1/\rho$ is a fractional certificate quantity [10]; we do not introduce it.

3 WHEN IS LEARNING THE FORMAT FREE?

Proposition 2 (Unknown-rate oracle). *The oracle coefficient is* $K_{\mathrm{or}} = 1/\psi_p(\rho)$. *Every optimal oracle allocation uses only the true better formats.*

Proof. Set $\lambda = (2q-1)/(2p-1) < 1$. A binary symmetric contraction maps p to q and any candidate $u \geq 1/2$ to $v = 1/2 + \lambda(u-1/2)$, including sign complements. Data processing bounds each worse-format divergence by its better-format counterpart. Aggregate both formats' weights by coordinate, choose a least-covered F, and fit $u = \max\{1/2, p-(2p-1)w(F)\}$. Feasibility requires $w(F) > 0$, since an unqueried wrong set is indistinguishable at the true rates. The resulting information is at most $\psi_p(w(F)) \leq \psi_p(\rho)$. Better-only sampling at cover-optimal weights attains the bound. Any positive worse-format mass makes the contraction strict since $u < p$, excluding it at equality. □

For $A \subseteq [G]$ and $F \in \mathcal{H}$, define $W_A = \sum_{g\in A} w(S_g)$, $T_A = \sum_{g\in A} w(F\cap S_g)$, $t = w(F)$, and

$$\begin{aligned} E_p(w;F,A) &= W_A\psi_p(T_A/W_A) \\ &\quad + (1-W_A)\psi_p((t-T_A)/(1-W_A)), \end{aligned} \tag{5}$$

$$J_p(w) = \min_{F\in\mathcal{H},\, A\subseteq[G]} E_p(w;F,A), \tag{6}$$

with zero-mass blocks contributing zero.

Theorem 3 (Catalogue–sharing boundary). *For every allowed* $q < p$,

$$K = K_{\mathrm{or}} \iff \exists w \in \Delta_d : J_p(w) \geq \psi_p(\rho). \tag{7}$$

Failure of the criterion implies a strictly positive premium. The equality boundary is independent of q.

Proof. For better-only weights, a group pool of mass W and reversed fraction s contributes $W[(1-s)\,\mathrm{kl}(p,u) + s\,\mathrm{kl}(p,1-u)]$. Minimizing over $u \geq 1/2$ gives $W\psi_p(s)$, attained at $\max\{1/2, p-(2p-1)s\} \leq p$. A candidate type pattern splits groups into two pools sharing two rates. Fit the pools independently and assign the larger fitted rate to p'; both orientations are allowed. Thus (6) is exactly their least alternative information. By Proposition 2, equality requires a better-only oracle optimum. Scaling its weights by $1/\psi_p(\rho)$ is feasible for (2) exactly when (7) holds. Conversely such a witness attains the oracle bound. Closed information constraints and compact bounded-cost allocation sets exclude unattained limiting equality. Only p, not q, enters this test. □

This is convex feasibility: each entropy perspective in (5) is concave in w. For fixed F, w, sort used groups by $w(F\cap S_g)/w(S_g)$; a minimizing two-pool assignment is a prefix cut, since the rate-assignment cost difference is affine and monotone in this fraction. Hence evaluating $J_p(w)$ uses $O(|\mathcal{H}|G\log G)$ operations after mass preprocessing. The flip family may be exponential; this is not a polynomial-time catalogue algorithm. Outside equality, an optimal learner may use both formats, and $1/\max_w J_p(w)$ need not equal K.

Corollary 4 (A unique oracle allocation). *Suppose* $\rho < 1/2$, *and the cover-optimal coordinate allocation* w^* *is unique. Then* $K = K_{\mathrm{or}}$ *exactly when* $\mathrm{supp}(w^*)$ *lies in one sharing group.*

Proof. Any equality witness lies in $\mathcal{W} = \{w \in \Delta_d : w(F) \geq \rho\ \forall F\}$. For binding F, strict entropy concavity below $1/2$ forces $w(F\cap S_g) = \rho w(S_g)$ for every used group; otherwise isolating that group reduces (5) below $\psi_p(\rho)$. Each normalized group restriction $v_g = w|_{S_g}/w(S_g)$ preserves all binding constraints. Small positive and negative moves along $v_g - w$ preserve the slack constraints and support. Uniqueness therefore forces $v_g = w$, possible only with one used group. Conversely a cover-optimal allocation in one group attains (7). □

Why group sizes do not suffice. Take $U_0 = 0$, $U_1 = 3\theta_1 + 2\theta_2 + 2\theta_3 + \theta_4 + \theta_5$ at $\theta = (+1)^5$. All cube winners are unique, and $\mathcal{H} = \{12, 13, 234, 235, 145\}$, with $12 = \{1,2\}$, etc. Primal weights $(3,2,2,1,1)/9$ and dual set weights $(1,1,2,2,3)/9$ certify $\rho = 5/9$, so $K_{\mathrm{or}} = 1/\mathrm{kl}(p,1/2)$. In this saturated regime equality requires some group to support weights covering every F by at least $1/2$: otherwise isolating a group with a smaller reversed fraction strictly lowers its entropy cut. For groups $(12, 345)$, uniform weight on 12 suffices. For $(14, 235)$, the restricted cover values are 0 and $1/3$, so the premium is positive. Both partitions have sizes $(2,3)$.

4 A CERTIFIED ACQUISITION LOOP

The implementation maintains the joint likelihood of preferences, group formats and continuous rates. It enumerates the finite sign/type states and maximizes the two ordered rates analytically

for each state. From the resulting joint maximum-likelihood state, it forms the minimal wrong-item sets and solves (2) numerically. Dividing the allocation by its total gives a query distribution $\widehat{w}_t$; an unidentified rate gap gives a uniform target instead. Evidence is retained and early type estimates are revised.

For concreteness, index the $2d$ actions by $e = 0, \dots, 2d-1$. At zero-indexed round $t = 2dn^2 + s$, $0 \le s < 2d$, force action s; at other rounds choose an action maximizing

$$\sum_{u=0}^{t} \widehat{w}_u(e) - N_e(t).$$

Targets are refreshed after $\max\{2d, \lceil\sqrt{t+1}\rceil\}$ further replies. The evaluated solver uses tolerance 10^{-6} and at most 150 iterations. This is a finite implementation of information tracking; the ideal coefficient claim instead requires forced-only estimates, vanishing actual numerical errors and the associated summable-tail argument.

Acquisition and certification have different roles. Set $\ell_t(i) = \sup_{h:f(\theta)=i} \log L_t(h)$ and choose a maximizing item $\widehat{i}_t$. Return it only if

$$\ell_t(\widehat{i}_t) - \max_{i \neq \widehat{i}_t} \ell_t(i) > \log(1/\delta) + \sum_e \log(N_e(t)+1). \tag{8}$$

The threshold gives anytime error control under any predictable acquisition policy. To see this, integrate each action's unrestricted Bernoulli parameter against a uniform Beta prior. Its saturated likelihood is at most $N_e + 1$ times its mixture likelihood. On a wrong certificate the true state's likelihood is in a competing item class, so (8) implies that the mixture likelihood ratio exceeds $1/\delta$. Ville's inequality bounds this event by δ. The evaluated gate uses exact rational maximum-likelihood fits and outward log intervals; overlapping intervals mean continue, or abstain at the cap.

This separates recommendation certification from allocation efficiency. The prototype's all-data maximum-likelihood estimates and fixed numerical tolerances differ from the ideal forced-sample, vanishing-error tracking contract. The certificate remains valid independently of allocation approximations; a numerical failure yields no certified recommendation.

5 CONTROLLED MECHANISM CHECKS

Four predeclared cells pair the weighted five-coordinate catalogue under the size-matched partitions above, and a six-coordinate three-item catalogue under one group versus $(123, 456)$. For the latter, the unique cover weights and value are

$$w^* = (3,2,2,1,1,4)/13, \qquad \rho = 5/13.$$

Corollary 4 predicts zero versus positive premium. Its utilities are

$$\begin{aligned} U_0 &= 0, \\ U_1 &= -6\theta_1 - 4\theta_2 - 4\theta_3 - 2\theta_4 - 2\theta_5, \\ U_2 &= -1 - 2\theta_2 - 2\theta_3 - 4\theta_4 - 4\theta_5 - 8\theta_6. \end{aligned}$$

Exact rational checks verify all 96 cube winners, flip families and cover witnesses across the two catalogues.

We fix $p = .8, q = .6, \theta = (+1)^d$ and $z_g = 0$ for all groups; thus the partition contrast changes sharing information alone. A fixed format can work well on these particular anchors. Six independent response tapes per catalogue are paired across partitions, three allocators and $R = \log(1/\delta) \in \{3, 9\}$. Decision-KL, repaired item-information-gain (IG) and round-robin share forced exploration, costs, exact-gate checkpoints and a 2,500-query cap. IG integrates the joint state with order-six quadrature and relative numerical ties. Protocol, seeds and analysis were frozen before main outcomes; no path was replaced or used for tuning.

Table 1: Mean query cost on six tapes per row (↓). Failed paths receive cost 2,500; these restricted costs are not completed stopping times. †: one numerical failure among six paths.

Catalogue / partition	R	KL	IG	RR
Weighted5 / zero	3	182.3	224.7	533.3
	9	227.7	295.2	581.3
Weighted5 / positive	3	222.0	235.0	449.7
	9	258.2	267.2	534.3
Three-item / zero	3	232.2	189.8	443.5
	9	264.0	216.7	536.2
Three-item / positive	3	694.8†	308.8	513.2
	9	758.7†	382.3	588.3

Across 144 paths, 142 certified with no observed wrong recommendation, zero reached the query cap, and two KL paths failed at the same paired seed's 213-query update, once per R. All raw answers and 142 terminal certificates replay correctly. The analytic risk bound e^{-R} follows from the model and gate, not the zero observed error count. The two failures remain in Table 1; the current numerical allocator is not robust on every tested path.

Sharing produces a clear finite effect in the three-item example: at $R = 9$, splitting the group increases IG cost by 165.7 queries (paired tape-bootstrap 95% interval $[143.5, 187.5]$). The weighted-five example gives -28.0 ($[-117.8, 35.0]$), and round-robin even reverses the ordering. These observations do not contradict the theorem: it compares optimal unknown-type and known-type coefficients within an instance, not arbitrary policies across partitions at finite confidence. No empirical oracle was run. KL is cheaper on the weighted-five anchors, but IG is stronger on the three-item anchors; there is no overall KL superiority claim. Intervals resample the six tapes within each fixed catalogue, preserving all pairing; they do not measure generalization across users or catalogues.

Scope. Our results concern fixed preferences, known groups and two globally shared symmetric iid reliability levels. They do not establish such sharing in natural dialogue, correlated repeated responses, changing preferences or unknown groups. The catalogue examples isolate an information-cost mechanism; they are not a benchmark of user satisfaction or Web deployment. Dedicated format reports or substantial labelled calibration can change the boundary. Learning valid groups and validating the response model remain necessary before applying the criterion to people or LLM-simulated users.

6 CONCLUSION

Learning how to ask is free at first order only when the decision evidence can absorb uncertainty about the question channel. For

recommendation, that property belongs to the catalogue and sharing partition jointly. The resulting criterion identifies when query-format adaptation is a genuine information burden and supplies a decision-directed, certifiable interface for studying it.

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